Real traces

Mooncake conversation (chat)

Mooncake toolagent (tools)

FreeInference (agent)

Trace-driven simulator

one request at a time

CPU tier + GPU recompute: measured on H200

SSD / NFS class: throttled emulation (constants from an earlier platform)

2 **return-time foresight**

MPS trade-off curve

Measured on the GPU

one H200, real runs

cost constants from microbenchmarks

1 3 **write, restore time**

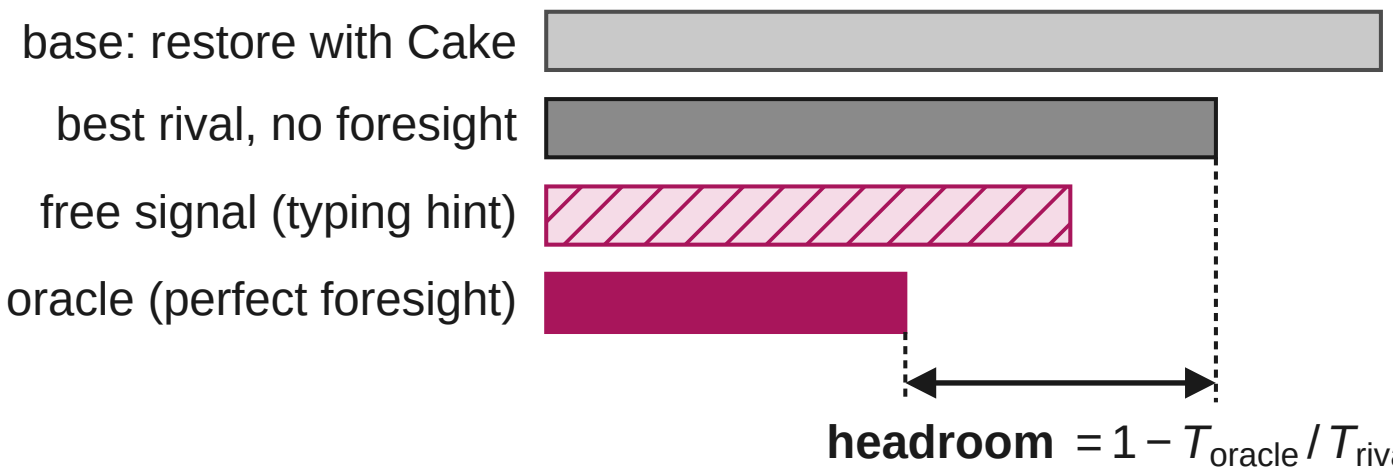
INT4 on vs. off

MPS microbench + check

scaled-down replay: measured / simulated TTFT 0.911–0.987, same ranking

Per decision: four arms on the same trace

bar = mean TTFT of returning requests (schematic)



Preregistered threshold: worth it if the bound is ≥15%